

Cleaning Audit Report

Dataset: adult_sample.csv · Session: 20260530-062547-f0fd71 · 2026-05-30 06:25 UTC

Summary

Rows (in → out)	6,000 → 5,878
Columns (in → out)	15 → 21
Total decisions logged	38
Automated (mechanical)	22
Human-confirmed / default	16
Analyst overrides	0

Cleaning Flowchart



What Changed

Format resolution

6 column name(s) standardised; 6 column(s) re-typed to their correct data type; 3 column(s) had placeholder tokens (e.g. '?') converted to missing; 10 other mechanical fix(es) applied.

Duplicate detection

24 duplicate row(s) removed.

Missingness

98 rows removed; 2 variable(s) imputed or flagged.

- workclass: 365 missing (6.1%) — Fill the gaps with the most common value
- occupation: 367 missing (6.1%) — Fill the gaps with the most common value
- native_country: 98 missing (1.6%) — Delete the rows with a missing value

Outlier detection

No flagged outliers acted on.

- age: 29 value(s) flagged — Keep these values — they are valid
- fnlwgt: 154 value(s) flagged — Leave for now — needs a closer look
- education_num: 217 value(s) flagged — Keep these values — they are valid
- capital_gain: 469 value(s) flagged — Leave for now — needs a closer look
- capital_loss: 293 value(s) flagged — Leave for now — needs a closer look
- hours_per_week: 1,634 value(s) flagged — Keep these values — they are valid

Feature engineering

6 transformation(s) applied.

- capital_gain: Bin into quantile groups → `capital\_gain\_bin`
- capital_loss: Bin into quantile groups → `capital\_loss\_bin`
- fnlwgt: Robust-scale (median / IQR) → `fnlwgt\_scaled`
- capital_gain: Log-transform (log1p) → `capital\_gain\_scaled`
- capital_loss: Log-transform (log1p) → `capital\_loss\_scaled`
- native_country: Frequency encode → `native\_country\_freq`

Variable Summary

Variable	Type	Missingness	What happened
age	integer	0.0%	Re-typed; Keep these values — they are valid
workclass	categorical	6.1% → 0.0%	Trimmed; Converted; Fill the gaps with the most common value

fnlwgt	integer	0.0%	Re-typed; Leave for now — needs a closer look; Robust-scale (median / IQR)
education	categorical	0.0%	Trimmed
education-num → education_num	integer	0.0%	renamed; Re-typed; Keep these values — they are valid
marital-status → marital_status	categorical	0.0%	renamed; Trimmed
occupation	categorical	6.1% → 0.0%	Trimmed; Converted; Fill the gaps with the most common value
relationship	categorical	0.0%	Trimmed
race	categorical	0.0%	Trimmed
sex	categorical	0.0%	Trimmed
capital-gain → capital_gain	integer	0.0%	renamed; Re-typed; Leave for now — needs a closer look; Bin into quantile groups; Log-transform (log1p)
capital-loss → capital_loss	integer	0.0%	renamed; Re-typed; Leave for now — needs a closer look; Bin into quantile groups; Log-transform (log1p)
hours-per-week → hours_per_week	integer	0.0%	renamed; Re-typed; Keep these values — they are valid
native-country → native_country	categorical	1.6% → 0.0%	renamed; Trimmed; Converted; Delete the rows with a missing value; Frequency encode
income	categorical	0.0%	Trimmed
capital_gain_bin	categorical	0.0%	derived (new column)
capital_loss_bin	categorical	0.0%	derived (new column)
fnlwgt_scaled	numeric	0.0%	derived (new column)
capital_gain_scaled	numeric	0.0%	derived (new column)
capital_loss_scaled	numeric	0.0%	derived (new column)
native_country_freq	numeric	0.0%	derived (new column)

Final Dataset Profile

age

Type: integer · Missing: 0.0% (0/5,878) · Unique: 69

Mean	Median	Std	Min	Max	Q25	Q75
38.77	37.00	13.76	17.00	90.00	28.00	48.00

workclass

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 8

Value	Count
Private	4,441
Self-emp-not-inc	471
Local-gov	395
State-gov	208
Self-emp-inc	193

fnlwgt

Type: integer · Missing: 0.0% (0/5,878) · Unique: 5,309

Mean	Median	Std	Min	Max	Q25	Q75
188,069	176,982	103,957	14,878	1,455,435	117,008	237,617

education

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 16

Value	Count
HS-grad	1,949
Some-college	1,358
Bachelors	964
Masters	282
Assoc-voc	237

education_num

Type: integer · Missing: 0.0% (0/5,878) · Unique: 16

Mean	Median	Std	Min	Max	Q25	Q75
10.02	10.00	2.54	1.00	16.00	9.00	12.00

marital_status

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 7

Value	Count
Married-civ-spouse	2,692

Never-married	1,915
Divorced	804
Widowed	203
Separated	193

occupation

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 14

Value	Count
Craft-repair	1,122
Exec-managerial	736
Prof-specialty	709
Adm-clerical	679
Sales	661

relationship

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 6

Value	Count
Husband	2,374
Not-in-family	1,547
Own-child	886
Unmarried	628
Wife	275

race

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 5

Value	Count
White	5,045
Black	573
Asian-Pac-Islander	164
Amer-Indian-Eskimo	51
Other	45

sex

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 2

Value	Count
Male	3,898
Female	1,980

capital_gain

Type: integer · Missing: 0.0% (0/5,878) · Unique: 85

Mean	Median	Std	Min	Max	Q25	Q75
989.32	0.0000	7,038	0.0000	99,999	0.0000	0.0000

capital_loss

Type: integer · Missing: 0.0% (0/5,878) · Unique: 64

Mean	Median	Std	Min	Max	Q25	Q75
92.70	0.0000	413.54	0.0000	3,900	0.0000	0.0000

hours_per_week

Type: integer · Missing: 0.0% (0/5,878) · Unique: 79

Mean	Median	Std	Min	Max	Q25	Q75
40.46	40.00	12.35	1.00	99.00	40.00	45.00

native_country

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 40

Value	Count
United-States	5,386
Mexico	103
Philippines	35
Canada	26
Cuba	24

income

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 2

Value	Count
<=50K	4,482
>50K	1,396

capital_gain_bin

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 1

Value	Count
all	5,878

capital_loss_bin

Type: categorical · Missing: 0.0% (0/5,878) · Unique: 1

Value	Count
all	5,878

fnlwgt_scaled

Type: numeric · Missing: 0.0% (0/5,878) · Unique: 5,309

Mean	Median	Std	Min	Max	Q25	Q75
0.0919	0.0000	0.8619	-1.34	10.60	-0.4973	0.5027

capital_gain_scaled

Type: numeric · Missing: 0.0% (0/5,878) · Unique: 85

Mean	Median	Std	Min	Max	Q25	Q75
0.7008	0.0000	2.40	0.0000	11.51	0.0000	0.0000

capital_loss_scaled

Type: numeric · Missing: 0.0% (0/5,878) · Unique: 64

Mean	Median	Std	Min	Max	Q25	Q75
0.3740	0.0000	1.63	0.0000	8.27	0.0000	0.0000

native_country_freq

Type: numeric · Missing: 0.0% (0/5,878) · Unique: 24

Mean	Median	Std	Min	Max	Q25	Q75
0.8401	0.9163	0.2522	0.0002	0.9163	0.9163	0.9163

Methods

The dataset `adult\_sample.csv` entered the pipeline with 6,000 rows and 15 variables. 122 rows were removed during cleaning, leaving 5,878 rows and 21 variables. Of 38 logged decisions, 22 were applied automatically and 16 required a judgment call. Every decision — automated or human — is captured in the accompanying `decisions.json`, and `clean\_script.py` replays the entire process from the raw input.

Generated by DataClean Agent. Every transformation is reproducible from the accompanying clean_script.py.